

COTEI RUXANDRA-MARIA

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Education

National College “Nicolae Grigorescu” • Câmpina, România • BAC Diploma •
Mathematics and Computer Science • GPA: 9.88 September 2018 - June 2022
Technical University of Cluj-Napoca • Cluj-Napoca, România • Bachelor’s Degree in
Computer Science and Engineering October 2022 - Present

Work Experience

CodeRun • Cluj-Napoca, România • Participated in the 3rd edition. 2022
International Collegiate Programming Contest (ICPC) • Cluj-Napoca, România •
Participating for three years, reaching countrywide level. 2022 – Present
Cloudflight Coding Contest (CCC) • Cluj-Napoca, România • Participated for two
editions. 2023 – 2024
Women In Data Science (WIDS) • Cluj-Napoca, România • Participating this year for the
first time. 2025 – Present

Volunteer Work

Board of European Students of Technology (BEST) • Cluj-Napoca, România
• helping organize a job fair with 1200+ participants 2023 – Present
• overseeing coding contest participants for two edition
• conducting interviews and assessing individuals as part of the recruiting process

Projects

DX Ball — Assembly April 2023
• Developed a brick-breaker game using Assembly, implementing collision detection, paddle movement, and score tracking for an optimized gaming experience.
ATM Machine — VHDL June 2023
• Designed an ATM simulation with hardware methods, featuring user authentication, balance inquiry, cash withdrawal using a greedy approach, and a deposit function through a hardware-based implementation.
MIPS Processor — VHDL April 2024
• Built a MIPS processor from scratch in VHDL, implementing both single-cycle and pipelined versions. Included hazard detection, forwarding, and stage coordination for efficient instruction execution.
Pac-Man AI — Python October 2024
• Implemented an AI-powered Pac-Man game using pathfinding algorithms to navigate the maze and evade or chase ghosts dynamically.
Snake Game — Arduino, Processing November 2024
• Developed a Snake game using Arduino for hardware input and Processing for graphics. Implemented real-time control via serial communication and increasing gameplay difficulty.

OpenGL Concert Rendering — C++

December 2024

- Created a 3D concert visualization using OpenGL, simulating stage lighting, audience interaction, and real-time animation effects.

RSA Encryption — VHDL, C++

January 2025

- Implemented RSA encryption using VHDL for hardware-based security and C++ for encryption and decryption of messages.

Technical Skills

Experienced languages: C++, Python, VHDL, Java, C, Haskell

Other languages: Assembly, Elm, SQL, HTML, CSS, JavaScript

Frameworks/Technologies: Linux, React, Arduino, Processing, Git/GitHub, Docker

Language Skills

Romanian – Native

English – Advanced

French – Intermediate